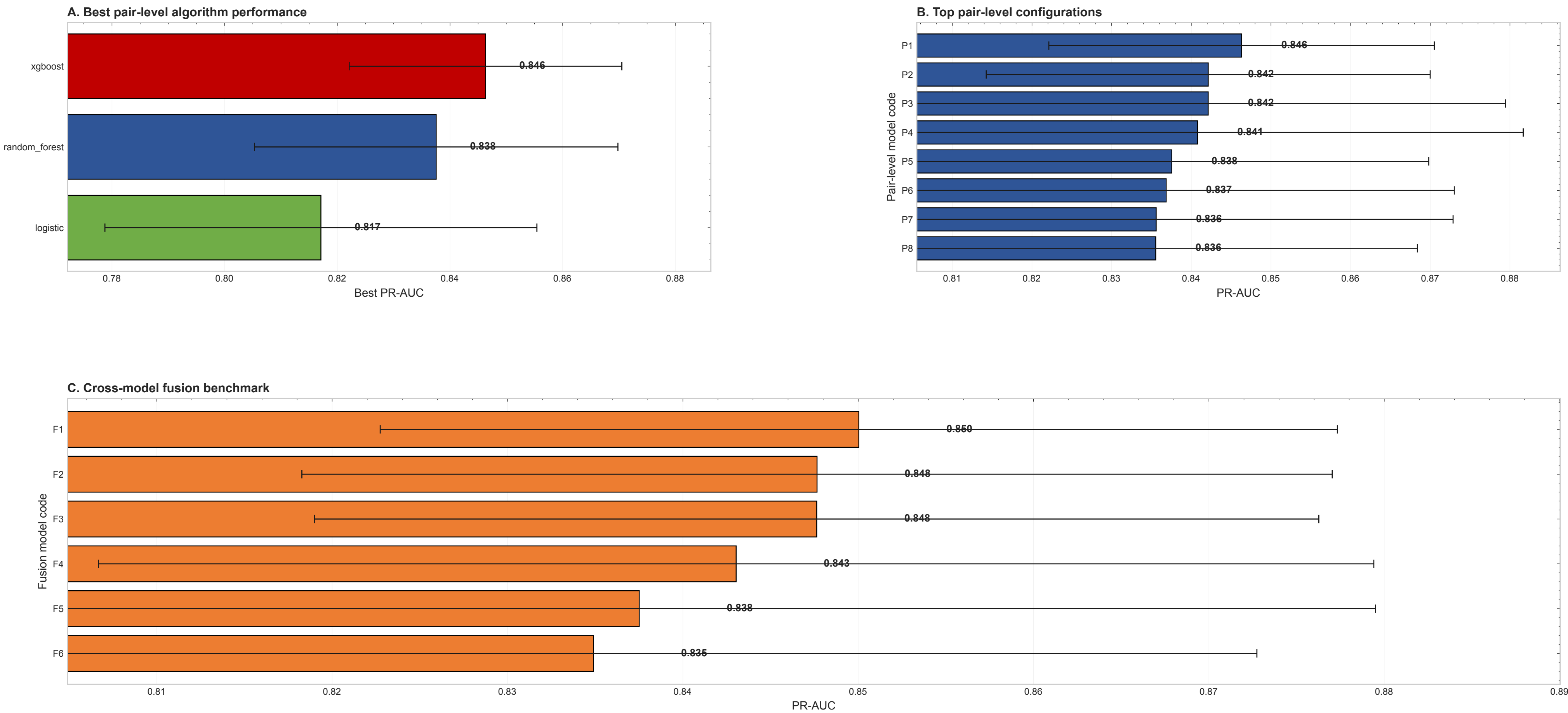


Figure 4. Pair-level machine learning and cross-model fusion benchmarks



D. Model code mapping and performance summary					
Code	Group	Model configuration	ROC-AUC	PR-AUC	Brier
P1	Pair-level ML	ESM2-3B fusion xgboost	0.759	0.846	0.215
P2	Pair-level ML	ESM2-3B per_sequence xgboost	0.757	0.842	0.217
P3	Pair-level ML	ESM2-650M per_sequence xgboost	0.758	0.842	0.215
P4	Pair-level ML	ESM2-650M fusion xgboost	0.756	0.841	0.216
P5	Pair-level ML	ESM2-3B per_sequence random_forest	0.754	0.838	0.204
P6	Pair-level ML	ESM2-3B fusion random_forest	0.755	0.837	0.204
P7	Pair-level ML	ESM2-150M per_sequence xgboost	0.754	0.836	0.217
P8	Pair-level ML	ProtBERT-BFD fusion xgboost	0.751	0.836	0.218
F1	Cross-model fusion	ESM650 + ESM3B + ProtBERT-BFD xgboost	0.766	0.850	0.212
F2	Cross-model fusion	ESM3B + ProtBERT-BFD xgboost	0.762	0.848	0.216
F3	Cross-model fusion	ESM650 + ESM3B xgboost	0.764	0.848	0.212
F4	Cross-model fusion	ESM650 + ProtBERT-BFD xgboost	0.757	0.843	0.217
F5	Cross-model fusion	ESM3B + ProtBERT-BFD random_forest	0.754	0.838	0.203
F6	Cross-model fusion	ESM650 + ESM3B random_forest	0.753	0.835	0.203